

Part II: Say True or False Questions (0.5 each: $0.5 \times 12 = 6$ points)

1. Related with body fluid and transportation

- ☐ ~~F~~ a. Hydrated potassium couldn't pass through its channel
- ☐ ~~F~~ b. Capillary walls are highly permeable to proteins
- ☐ ~~F~~ ☒ c. More sodium is found in the tissue than in the blood
- ☒ ~~F~~ d. Precapillary relaxation reduces tissue fluid accumulation
- ☒ ~~T~~ e. More water is found in the human tissues than in the blood vessels
- ☒ ~~T~~ f. The more the fat in the body is the less the water
- ☒ ~~F~~ g. Fluid returns back into the circulation from tissue only through veins

2. Related with the body fluid tonicity and related disorder

- ☒ ~~T~~ a. High antidiuretic hormones increases blood tonicity
- ☒ ~~F~~ b. More proteins in the plasma aggravate edema
- ☒ ~~T~~ c. profuse sweating brings hypernatremic dehydration
- ☒ ~~T~~ d. Dehydration increases hemoglobin concentration.
- ☒ ~~T~~ e. No osmotic gradient formation in the isotonic dehydration.

Part III: Matching (0.5 each: $6 \times 0.5 = 3$)

1. match the organelle under column A with the appropriate functions under column B and write the number on from column B on the space under column A

<u>Column A</u>	<u>Column B</u>
<u>A.</u> Lysosomes	1. Synthesis of proteins
<u>B.</u> Golgi apparatus	2. Complete proteins synthesis
<u>C.</u> Peroxisomes	3. Aerobic metabolism
<u>D.</u> Smooth endoplasmic reticulum	4. Neutralize hydrogen peroxide
<u>E.</u> Bounded ribosomes	5. Intracellular digestion
<u>F.</u> Mitochondria	6. Lipid synthesis
	7. Control cell activities

A. O-Glycosidic linkage

B. Ester bond

C. N-Glycosidic linkage

D. Disulfide bond

37. Among the RNA types, the smallest in size and also forms secondary structure is:

A. messenger RNA B. transfer RNA C. small nuclear RNA D. ribosomal RNA

38. Which of the following is more risky of having children with chromosomal abnormalities?

A. Longer diplotene stage of primary oocyte in prophase of meiosis I

B. The resting stage of secondary oocyte at metaphase

C. Diplotene stage of primary oocyte in metaphase of meiosis I

D. Short diplotene stage of primary oocyte following puberty

E. None of the above

39. Which of the following is caused by microdeletion of chromosomes?

A. Mosaicism B. Trisomy 13 C. Angleman syndrome

D. Down syndrome E. Klinefelter disease

40. Which of the following organs serves as storage and maturation of a spermatocyte?

A. Semineferous tubules B. Straight tubules C. Epididymis D. ductus deferens

E. Spermatic cord

41. Which of the following is not a result of meiotic division?

A. Spermatid B. Primary spermatocyte C. Secondary spermatocyte

D. Polar body E. None of the above

42. The _____ lies superficial to the plasma membrane of the oocyte in the matured ovum?

A. Granulosa cells B. Follicular cells C. Zona pellucida

D. Theca externa E. Antrum

43. Primordium means, the embryo and its associated membranes. — *conceptus*

A. True B. False

44. Wrong statement about the crossover occurring during meiosis?

A. An interchange of chromatid segments between paired homologous chromosomes

B. It enhances genetic variability

C. It occurs at Meiotic

D. A chiasma is seen at the point of interchange

E. None of the above

45. Which of the following contains diploid number of 46 chromosomes?

A. Polar bodies B. Spermatids C. Primary oocyte

D. An ovum E. Spermatozoa

26. Which of the following is not polymer of α -D-glucose?
 A. Cellulose B. amylase C. glycogen D. amylopectin
27. Which amino acid side chain can form hydrogen bonding during protein structure formation?
 A. Alanine B. Leucine C. Tyrosine D. Tryptophan
28. Tyrosine and serine are similar by having side chain of:
 A. aromatic amino acids B. hydroxyl amino acids C. aliphatic groups
 D. acidic side chains E. Nonpolar side chains
29. Desmosine, found in the fibrous protein elastin, is derived from:
 A. Arginine B. Proline C. Histidine D. Lysine E. Leucine
30. S-adenosyl methionine is an amino acid derivative required in reactions which involve:
 A. transacetylation B. uridylation C. transmethylation D. transamination E. adenylation
31. The ribosome is classified under
 A. Lipoproteins B. Nucleoproteins C. Glycoproteins
 D. Flavoproteins E. Hemoglobin
32. Which one of the following proteins has a different shape than the others:
 A. Immunoglobulin B. Keratin C. Elastin D. Myosin E. Fibrinogen
33. The roles of molecular chaperones include the following except which?
 A. They shield interactive surface of non-native polypeptide
 B. Mostly expressed constitutively
 C. Recognize hydrophilic residues
 D. They can reverse intramolecular misfolding
34. Which one is true about group II chaperonin system
 A. found in eukaryotic cytosol
 B. 60 KD heat shock protein family
 C. contains monomeric units
 D. Energy independent process
35. Consequence of protein denaturing agents include:
 A. Increase in the chance of aggregation
 B. Decrease in solubility
 C. Increase in exposed hydrophobic side chains
 D. All of these
36. The type of bond between the sugar and phosphate group that occurs during the formation of DNA strands is:

E. None of the above

16. Which one of the following does not describe water?

- A. Symmetrical electric charge distribution B. dipolar structure
C. ability to interact with polar and charged molecules D. principal component of cells

17. Aldehydes and ketones contain _____ functional group

- A. carboxyl B. amine C. carbonyl D. Alcohol

18. What do you conclude if you boil the urine of a diabetic patient with a Benedict's reagent (blue color) and find the of the reagent changed to yellowish-green?

- A. Likely excretion of glucose in urine B. definite presence of glucose in urine
C. likely presence of sucrose in urine D. likely presence of oxidizing agent in urine

19. Which of the following is the relatively weakest non-covalent interaction?

- A. Hydrogen bond B. hydrophobic interaction
C. electrostatic interaction D. all the above mentioned have equal bond strength

20. Which interaction occurs in an environment that excludes water?

- A. H-bonding B. salt bridge
C. hydrophobic interaction D. electrostatic interaction

21. Which of the following is a phospholipid antigen that can cause antibodies to occur in the bloodstream and is associated with human disease?

- A. Phosphatidylcholine B. phosphatidylserine
C. diphosphatidylglycerol (cardiolipin) D. phosphatidylinositol

22. Correct statement concerning H-bonding is/ are:

- A. Plays role in specific transfer of genetic information
B. plays role in specificity of genetic code
C. plays role in stabilizing structure of biologically important molecules
D. all are correct answers

23. Which of the following is an essential fatty acid likely obtained from dietary oils

- A. Palmitic acid B. linoleic acid C. oleic acid D. palmitoleic acid

24. An INCORRECT statement concerning sugar alcohols is:

- A. Aldehyde group is reduced B. ketone group is reduced
C. all carbon atoms contain hydroxyl groups D. aldehyde group is oxidized

25. An intracellular glycosaminoglycan (GAG) is:

- A. Chondroitin sulfate B. heparin C. keratan sulfate D. dermatan sulfate

- A. Osmole
B. Mole
C. Molecular weight
D. Type of the solute in the solution
E. None of the above
8. Which of the following electrolyte found in extracellular compartment in less amount than the others
A. Sodium
B. Chloride
C. Calcium
D. Bicarbonate
E. Phosphate
9. Which of the following ions is not influx in to the cell passively
A. Sodium
B. Calcium
C. Bicarbonate
D. Chloride
E. Potassium
10. All bring dehydration except one
A. More coffee
B. Excessive sweating
C. Strenuous exercise
D. Intravenous water infusion
E. More diuretic drugs
11. All of the following conditions facilitate antidiuretic secretion from the hypothalamus except one
A. Bleeding
B. Hypotonic solution infusion
C. Hypovolemia
D. Hyperosmolality
E. Ang II secretion
12. All of the following brings edema except one of the following
A. Oncotic pressure reduction x
B. Reduction of plasma proteins x
C. Increased capillary hydrostatic pressure x
D. Chronic heart disease x
E. None of the above
13. In which of the following age groups more water is found in their body
A. Infants
B. Fetus
C. Neonate
D. Children
E. Adults
14. If you have a patient with the liver or kiddies diseases, she/he may develop edema. This is because
A. Increase osmotic pressure
B. Increase plasma proteins x
C. Reduced osmotic pressure
D. Reduces hydrostatic pressure x
E. Reduces capillary permeability x
15. Which of the following is not involved in the regulation of body water
A. Hypothalamus
B. Kidney
C. Antidiuretic hormone
D. Angiotensin two

PART I. Choose the Most Appropriate Answer from the Given Alternatives.

1. Which of the following is true
 - A. Addition of hypotonic solution into ECF reduces cell volume
 - B. Hypertonic dehydration increase hematocrite unit of the blood -
 - C. Addition of hypotonic solution into ECF increase the osmolarity of the ECF
 - D. Addition of hypertonic solution to the ECF increases the cell volume
 - E. Watery diarrhea is the cause for hypotonic dehydration
2. Which of the following substance pass through the lipid bilayer by simple diffusion
 - A. Sodium
 - B. Potassium
 - C. Carbohydrate
 - D. Vitamin A
 - E. Bicarbonate ion
3. What is the osmotic pressure of a solution with 150 mmol/L of solution of CaCl_2
 - A. 5,790mmHg
 - B. 450mmHg
 - C. 38.6mmHg
 - D. 8,685mmHg
 - E. 5,500mmHg
4. Which of the following is true about the physiology of the cell membrane
 - A. Active ion channels are leaky
 - B. Active ion channels dissipate the potential differences
 - C. Sodium influx channels are active ion channels
 - D. Passive ion channels have gate
 - E. The center of the cell membrane is hydrophobic
5. All of the following conditions are very important for channels to select substances crossing the cell membrane except one
 - A. The size of the channels
 - B. The charge of the channels
 - C. The size of the substances
 - D. The binding site on the channels
 - E. The amount of water surrounding substances
6. What will be the amount of intracellular fluid volume and its concentration respectively if a person is infused with hypotonic solution
 - A. Increased and reduced
 - B. Reduced and increased
 - C. Reduced and reduced
 - D. Increased and increased
 - E. Not changed
7. Which of the following affects the osmotic pressure of a solution highly than the others

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